

National University of Computer and Emerging Sciences



CL1000- Introduction to Information and Communication Technologies

Number System Assignment (Hand Written)

Department of Computer Science

Instructions

Read the Instructions Carefully:

- Deadline for assignment submission is **23rd September,22 till 2:15 pm**
- Submit your own original work in hard copy **(hand written)**.
- **Late submissions** would not be entertained.
- Your title page must be containing **“Section_Name, Roll_Number, Your_Name, Subject_Name, and Instructor_ Name”**.
- It should be clear that in case of **copying**; all parties involved in the copy case will be graded **“Fail”**.

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Solve the Following Questions.

1. Convert $(10000002.0625)_{10}$ into an octal number and find its 8's complement.
2. Convert $(-1.001001100)_2$ into hexadecimal and decimal numbers.
3. Convert $(255.101)_8$ into decimal and hexadecimal numbers.
4. Convert $(ABCDEF.ABC)_{16}$ to octal and binary numbers, also find 7's complement of octal number and 2's complement of binary number.
5. Convert $(1F0C)_{16}$ to octal and then octal to decimal and then decimal to binary.
6. What will be the 1's and 2's complement of $(00101011010010000)_2$.
7. Find 10's complement of $(9981958)_{10}$ and convert the answer in Base 16 also find its 16's complement after that convert it into Base 2 also find its 1's and 2's complement.
8. Convert $[(((1010.1111)_2 - (10110.000)_2) + (79)_{10}) * (10)_2]$ into hexadecimal.
9. Perform a hexadecimal addition of $(B49C)_{16}$ and $(4E2F)_{16}$.
10. Express $(+235)_{10}$ and $(-235)_{10}$ into signed magnitude binary form.
11. Find the subtraction $(96258 - 43271)_{10}$ using the 10's-complement method.
12. What is the range of signed binary number. Also find the range of 8-bit signed number.
13. Convert floating pointer number (3.33) to binary number.
14. Convert -3.1061 to 32-bit single precision floating-point number.
15. Subtract $(11111111 - 100000000)_2$ using 2's complement.
16. Divide $(10101010)_2$ by $(1001100)_2$.

GOOD LUCK 😊